

Project: Gmail Inbox Cleanup via API — Filtering & Categorization

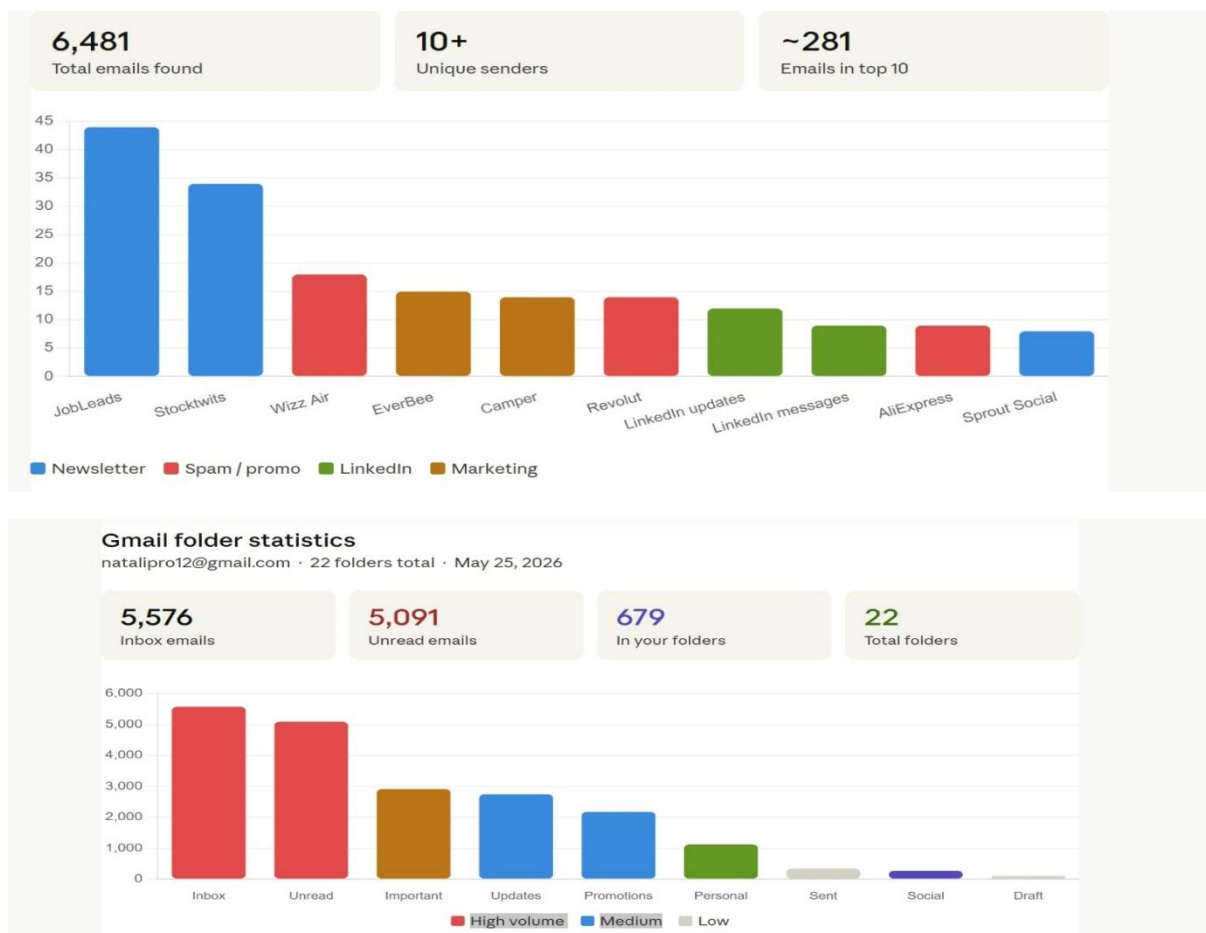
Objective

Reduce inbox clutter by connecting directly to a personal Gmail account via API, extracting message data at scale, and applying filtering/categorization logic to identify and remove low-value messages (spam, unsubscribe-pattern emails, promotional content).

Approach

1. **Data extraction** Connected to the Gmail account using the Gmail API to programmatically access inbox contents, rather than manually reviewing messages one by one.

1.Initial gmail box content



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Approach

1. **Data extraction** Connected to the Gmail account using the Gmail API to programmatically access inbox contents, rather than manually reviewing messages one by one.

2. **Filtering logic** Built rules to identify and flag low-value messages, including:

- Searching message bodies/subjects for patterns such as "unsubscribe" (a strong signal of promotional/marketing email)
- Identifying other common spam/marketing indicators (sender patterns, repeated senders, promotional keywords)

Statistic _ most common senders /spam



3. **Categorization** Grouped flagged messages into categories (e.g. promotional, spam, unsubscribe-pattern) to review before deletion, rather than deleting in one uncontrolled batch — ensuring important messages weren't accidentally caught by the filters.

4. **Outcome** Reduced total inbox volume from approximately **7,000 messages to 600** by systematically identifying and removing low-value email through the filtering/categorization logic above.

Skills Demonstrated

- API-based data extraction (Gmail API)
- Rule-based filtering and pattern matching at scale
- Categorization logic to reduce false positives before taking action
- Practical, self-directed problem solving using real-world data (not a synthetic dataset)

Relevance to Data Analyst Roles

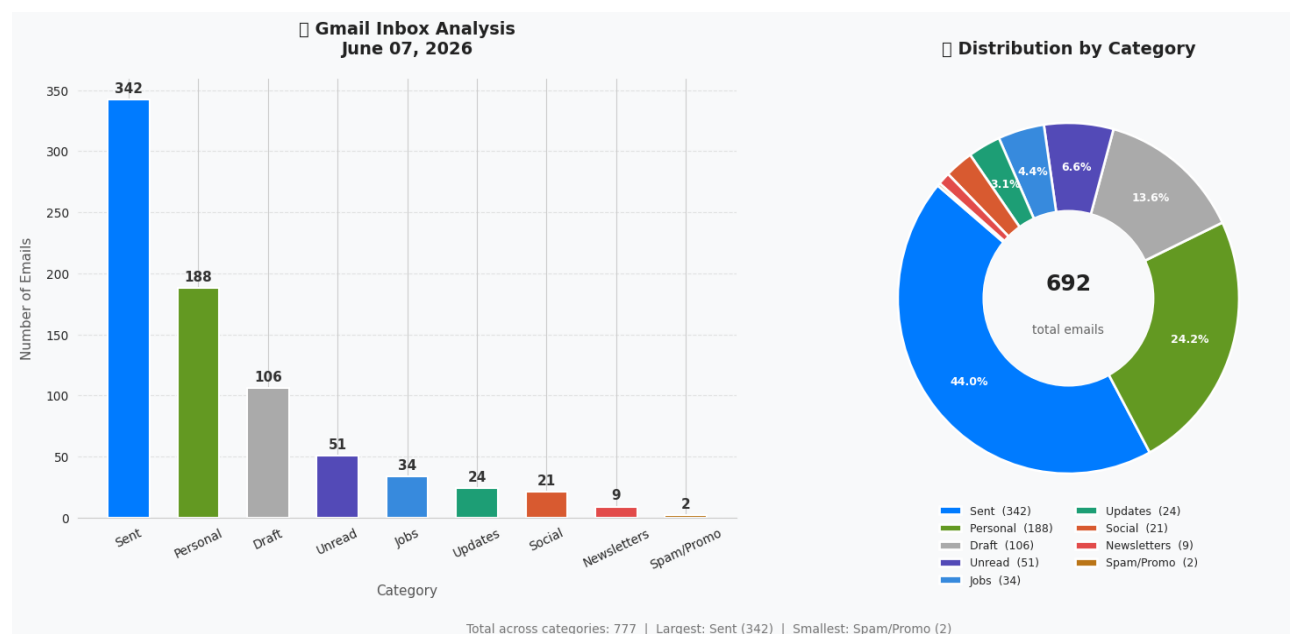
This project mirrors core responsibilities in data-driven investigation roles: extracting data via API, applying pattern-detection logic to a large volume of records, and separating signal (genuine spam/promotional content) from noise (legitimate messages) before taking action — directly comparable to distinguishing genuine quality issues from false positives in a content or translation review context.

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 STATS SUMMARY AFTER CLEANING

Sent	342 (44.0%)	Personal	188 (24.2%)	
Draft	106 (13.6%)	Unread	51 (6.6%)	
Jobs	34 (4.4%)	Updates	24 (3.1%)	
Social	21 (2.7%)	Newsletters	9 (1.2%)	Spam/Promo
(0.3%)				2

TOTAL 777

Inbox total 692

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